Meridian — Condensed Autopsy

A forensic measurement framework for RAG. The measurement layer is the contribution.

Thesis

RAG evaluation needs a deterministic trust anchor. LLM judges drift with model updates, temperature, and prompt wording. Meridian's Layer 1 classifies every retrieved span against ground-truth character offsets using pure set intersection — no embeddings, no model calls. Layer 2 (LLM-judged correctness and faithfulness) sits on top, pinned and separate. The layers never contaminate each other. When a number moves, the source is identifiable.

The proof: four silent measurement bugs were caught by the framework's verify-before-trust discipline before they shipped wrong numbers. Each was caught by instrument verification, not by the output looking wrong.

Headline numbers

Metric	Value	Claim type
Config-stack delta (correctness)	+7.9pp (60.5% → 68.3%)	Controlled (same index)
Config-stack delta (faithfulness)	+4.2pp (91.7% → 95.9%)	Controlled (same index)
Benchmark regime	4 corpora × 194q, 11,524-chunk combined index	—
Ruler calibration	3/4 corpora within ~2-3pp embedding-drift floor	Validated (Finding 44)
Measurement bugs caught	4 (each before shipping)	Methodology
Evidence trail	47 findings, 7 corrections	In docs/DECISIONS.md

Measurement-layer design

Layer 1 (deterministic, no LLM): Six failure types — DRM (wrong document), CBF (chunk boundary), SGP (span gap), ICR (wrong section), OVR (over-retrieval), OK (correct). Classification: `gt_chars & retrieved_chars` (character-set intersection). Per-span scoring (not merged-character-set) to handle multi-span evidence (43% of queries). P@k, R@k are character-overlap ratios.

Layer 2 (LLM-judged, separate): Correctness (span-informed: judge sees answer + golden evidence) and faithfulness (holistic groundedness: each claim vs full retrieved context, RAGAS definition). Pinned model (DeepSeek-v4-flash, T=0, thinking disabled). Never contaminates Layer 1.

Calibration: Paper's exact stack replicated (RCTS 500-char, text-embedding-3-large, dense-only, sqlite-vec). 3/4 corpora reproduce within ~2-3pp (embedding drift). ContractNLI diverges (benchmark-file provenance). Ruler confirmed (Finding 44).

The four caught bugs — post-mortem

Each bug produced a plausible-looking output indistinguishable from a correct result. Detection required checking the instrument, not the reading. This section is the thesis demonstrated.

Bug 1: Pro model-ID alias (Finding 34)

Aspect	Detail
Mechanism	PRO_MODEL="deepseek-chat" in run_headline.py. DeepSeek's legacy alias "deepseek-chat" routes to deepseek-v4-flash (documented f
Why invisible	The harness logged the REQUESTED model string (the constant), not the API response's served model field. Output records said model=
How caught	Billing dashboard: $0.00 on the deepseek-v4-pro line after a 776-query run. DeepSeek API docs confirmed the alias routing. Token/latenc
Fix	Model ID corrected to "deepseek-v4-pro". served_model field added to synthesize() (getattr(response._raw_response, 'model', None)). Co

Bug 2: BM25 channel mismatch (Finding 45)

Aspect	Detail
Mechanism	Combined-index dense retrieval (sqlite-vec) searched 4,496 mini-doc chunks. BM25 retrieval loaded the FULL corpus parquet (96,256 chu
Why invisible	ContractNLI and PrivacyQA completed with 0 errors and saved results. ContractNLI's mismatch (701 mini vs 3,797 full) was proportionally
How caught	CUAD crashed (96K-chunk BM25 index exceeded memory). Post-mortem traced the BM25 loading path: pd.read_parquet(params['parque
Fix	BM25 filtered to the same mini-doc set as the dense index before building the BM25Okapi index. Combined BM25: 11,524 chunks (chann

Bug 3: Zero-span offset (Finding 45)

Aspect	Detail
Mechanism	The combined sqlite-vec index was built by scrolling Qdrant collections and copying payloads. Qdrant payloads contain chunk_id, doc_id,
Why invisible	P@k/R@k are character-overlap ratios. With all spans at (0,0), the intersection with ground-truth spans was near-zero but not exactly zero
How caught	Sanity-check: CUAD R@8=0.049 with 100% routing recall and 8.0/8 correct-doc chunks contradicted expectations (perfect routing should
Fix	Span lookup from parquet files via chunk_id join. Spot-checked 3 chunks: looked-up span text matched saved content text exactly. P@k/F

Bug 4: Chunk-size inconsistency (Finding 46)

Aspect	Detail
Mechanism	MAUD's baseline SAC collection uses 2048-char chunks with ~512 overlap. ContractNLI, PrivacyQA, and CUAD use 512-char chunks wit
Why invisible	Per-corpus evaluations never compared chunk sizes across corpora. The combined-index headline mixed both sizes in one pool without f
How caught	External-baseline verification: comparing P@k to the paper's RCTS 500-char chunks. Investigation of whether the comparison was confou
Fix	Documented as Finding 46. CLAUDE.md corrected: "512-char (CNL/PQA/CUAD), 2048-char (MAUD)". External comparison: 3 corpora ur

Pattern: each bug produced a number that could have been reported as a finding. The Pro alias would have shipped "Pro adds nothing" (true, but for the wrong reason — it was flash-vs-flash). The span bug would have shipped CUAD R@8=0.049 as "retrieval barely works on CUAD" (false — real R@8=0.814). Verification caught the instrument error before it became a published conclusion.

Results

Config-stack delta (controlled, method-level)

Combined-index regime: 4 LegalBench-RAG corpora, 11,524 SAC chunks from 72 mini-split documents, 194 queries per corpus. Both arms use the same index, embedder (voyage-4), and judge.

Corpus	Correct Arm 0→1	Faith Arm 0→1	P@1 Arm 1	R@8 Arm 1	Latency
ContractNLI	62.9→71.1 (+8.2)	89.1→94.1 (+5.0)	0.422	0.810	5.6s
PrivacyQA	49.0→55.7 (+6.7)	94.4→97.9 (+3.5)	0.297	0.579	7.1s
CUAD	61.9→73.7 (+11.8)	92.1→96.0 (+3.9)	0.394	0.814	6.2s
MAUD	68.0→72.7 (+4.7)	91.3→95.5 (+4.2)	0.270	0.783	8.4s
Average	60.5→68.3 (+7.9)	91.7→95.9 (+4.2)	—	—	—

External comparison (system-vs-system)

vs arXiv 2408.10343 Table 5: RCTS 500-char, text-embedding-3-large, dense-only, no reranker. 3 un-confounded corpora (512-char ≈ 500-char). Full Meridian stack (SAC+CC+hybrid+routing+selector, voyage-4) vs bare baseline. The advantage bundles embedder + hybrid + fusion + routing — system-level, not method-alone.

Corpus	Meridian P@1	RCTS P@1	Meridian R@8	RCTS R@8	Note
ContractNLI	0.422	0.066	0.810	0.250	Benchmark provenance caveat
CUAD	0.394	0.020	0.814	0.317	—
PrivacyQA	0.297	0.144	0.579	0.424	—
MAUD	—	—	—	—	Excluded: 2048 vs 500-char confound

NFCorpus transfer (retrieval stack only)

BEIR medical IR (3,633 docs, 323 queries). nDCG@10 = 0.399 (CC α=0.1, routing OFF). Above classic baselines: BM25 0.325, BM25+CE 0.350, contriever 0.328. CC fusion transfers (+5.6pp over RRF). Span-forensic framework NOT exercised (BEIR has document-level relevance, no character spans). Routing correctly self-disabled (dispersed relevance). This is a retrieval-transfer result.

Routing boundary characterized: monotonically hurts on NFCorpus (OFF 0.397 > k=10 0.333 > k=5 0.273 > k=3 0.227). Routing is a concentrated-relevance technique — the sweep auto-detects when it helps vs hurts (Findings 23, 45, 47).

Limitations

- **Routing degrades to 76%** on ContractNLI combined regime (NDA/commercial-contract confusion). 47/194 queries get 0/8 correct-document chunks.
- **Routing hurts on dispersed relevance.** Confirmed on NFCorpus. Concentrated-relevance only.
- **Span framework requires character-span ground truth.** Does not apply to document-level benchmarks (most of BEIR, MS MARCO). Scope boundary.
- **MAUD external comparison confounded** (2048-char vs 500-char chunk granularity).
- **ContractNLI external baseline caveated** (benchmark-file provenance differs from paper's generation pipeline).
- **Affirmative-only evaluation.** Correctness = recall, not false-positive rate.
- **Synthesis variance ±2-4pp** (Finding 40). Sub-band deltas are noise.
- **External multipliers are system-level,** not method-level.

What Meridian is not

Not an autonomous research agent (v1 direction, abandoned). Not a SOTA-everywhere claim (beats classic baselines, not modern dense SOTA). Not a framework-transfer claim (span taxonomy ran on LegalBench-RAG only; NFCorpus was retrieval-transfer). Not a tutorial.

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47 findings with 7 corrections in docs/DECISIONS.md. Full methodology in docs/REPORT.md. Figures and reproducible generation in figures/.